BIO 005 Human Physiology · Week 13 notes · Dr. Sharilyn Rennie

Breathing and gas transport

Need-to-know respiratory anatomy, Boyle's law and the pressures of quiet and forced breathing, intrapleural pressure with pneumothorax and atelectasis, compliance, surfactant and airway resistance, lung volumes, spirometry patterns and dead space, partial pressures, diffusion and ventilation-perfusion matching, oxygen and carbon dioxide transport with the oxyhemoglobin curve and the Bohr and Haldane effects, oxygen content in anemia and carbon monoxide poisoning, the respiratory centers and chemoreceptors, exercise and altitude, the ventilatory response lab, and the respiratory terms. Week 13 competencies 1 to 22.

To print: these notes as a two-column PDF.

How to use these notes. Read these notes alongside your textbook reading for the week, then use them to review. Each part names where to read in Silverthorn and in OpenStax, and links Dr. Rennie's videos where there is one.

Your textbook: Silverthorn Chapter 18, Mechanics of Breathing, and Chapter 19, Gas Exchange and Transport. **Using OpenStax instead?** Anatomy and Physiology 2e, Chapter 22, The Respiratory System.

Need-to-know anatomy

Competency 1: Functions and functional zones.

In your textbook: Silverthorn Chapter 18, the parts on the functions of the respiratory system and the anatomy of the respiratory system.

Dr. Rennie's videos for this part: Functions of the respiratory system, Internal versus external respiration, Steps of respiration, 4 steps of collective respiration.

Respiration happens in four steps, and these notes follow them in order. This section covers only the structures you need to picture to follow the physiology.

The four steps of respiration

1. Ventilation (breathing): air moves between the atmosphere and the alveoli.
2. External respiration: O2 and CO2 are exchanged between alveolar air and the blood in the pulmonary capillaries.
3. Gas transport: the blood carries O2 to the tissues and CO2 back to the lungs.
4. Internal respiration: O2 and CO2 are exchanged between the blood in the systemic capillaries and the cells.

Cellular respiration, the use of O2 by mitochondria to make ATP, is a separate process inside cells. It is why the four steps are needed, and it is where the CO2 comes from.

What the respiratory system does

- Gas exchange: brings in O2 and removes CO2.
- pH regulation: changes blood pH by changing how much CO2 is breathed off.
- Protection: traps and removes inhaled particles and microbes.
- Vocalization: air moving past the vocal folds makes sound.
- Other jobs: smell, the respiratory pump that helps venous return, and angiotensin converting enzyme (ACE) on lung capillaries, which turns angiotensin I into angiotensin II.

Conducting zone and respiratory zone

	Conducting zone	Respiratory zone
Structures	Nasal cavity, pharynx, larynx, trachea, bronchi and bronchioles, down to the terminal bronchioles	Respiratory bronchioles, alveolar ducts, alveolar sacs and alveoli
Gas exchange	None	Yes, across the alveolar walls
Job	Carries air, and warms, humidifies and filters it	Exchanges O2 and CO2 with the blood
Wall	Cartilage in the trachea and bronchi; smooth muscle without cartilage in the bronchioles	A very thin epithelium wrapped in capillaries
Lining	Ciliated epithelium with mucus-secreting goblet cells in most of it	Type I and type II alveolar cells, with alveolar macrophages

Air reaching the alveoli is warmed to 37 °C (98.6 °F) and saturated with water vapor. Mucus traps particles, and cilia sweep it up to the pharynx to be swallowed (the mucociliary escalator). Cigarette smoke slows the cilia, so smokers cough to clear mucus.

Bronchi and bronchioles

The trachea, held open by C-shaped cartilage rings, splits into right and left primary bronchi, which branch again and again. Tubes narrower than about 1 mm, with no cartilage, are bronchioles. Their walls are mostly smooth muscle, so they can narrow and widen.

Alveoli

Tiny air sacs at the ends of the airways, hundreds of millions of them, with a total surface area of roughly 70 square meters.

Type I alveolar cells

Very thin, flat cells forming most of the alveolar wall. Gases diffuse across them.

Type II alveolar cells

Smaller, rounder cells that make surfactant and can divide to repair the wall. Alveolar macrophages roam the surface and eat particles and microbes.

Respiratory membrane

The barrier between alveolar air and blood: a type I cell, a fused basement membrane, and the capillary endothelium. In most places it is less than 1 µm thick.

The lungs, the pleura and the breathing muscles

Lungs

The right lung has three lobes and the left has two, leaving room for the heart. Lung tissue is elastic and always tends to recoil inward.

Pleural sacs

Each lung sits in its own closed sac. The visceral pleura covers the lung, and the parietal pleura lines the chest wall and diaphragm. A thin film of pleural fluid between them lets the layers slide but holds them together, like water between two panes of glass.

Diaphragm

A dome of skeletal muscle under the lungs and the main muscle of inspiration, supplied by the phrenic nerves (spinal cord levels C3 to C5).

External intercostals

Lift the ribs up and out during inspiration.

Internal intercostals

Pull the ribs down and in during forced expiration.

Accessory muscles of inspiration

The scalene and sternocleidomastoid muscles of the neck, which lift the upper ribs and sternum. The scalenes help a little even in quiet breathing and work much harder in forced inspiration, when the sternocleidomastoids join in.

Abdominal muscles

The rectus abdominis, external and internal obliques, and transversus abdominis, which push the abdominal organs up against the diaphragm in forced expiration.

All the breathing muscles are skeletal muscle with no rhythm of their own, so every breath depends on nerve signals, normally from the brainstem.

How air moves: Boyle's law and the breathing cycle

Competency 2: Pressure gradients and airflow.

In your textbook: Silverthorn Chapter 18, the parts on the gas laws and on ventilation, including inspiration and expiration.

Dr. Rennie's videos for this part: Pulmonary ventilation (normal ventilation), Pressure and volume (Boyle's law), Charles's law, Air flow and pressure, Air flow and pressure, syringe analogy, Requirements for ventilation, and quiet versus forced respiration, Describe pre-

inspiration (rest), Describe inspiration, Describe expiration, Muscles of respiration and air flow, part 1, Muscles of respiration and air flow, part 2, Non-thoracic effects of changes in abdominal pressure.

Air flows from higher to lower pressure, the same rule that moves blood: flow = pressure gradient ÷ resistance. The body cannot change atmospheric pressure, so it changes the pressure inside the alveoli by changing the volume of the chest.

Boyle's law

At a constant temperature, the pressure of a gas in a closed container is inversely related to its volume: P1 × V1 = P2 × V2. A bigger container means lower pressure; a smaller one means higher pressure.

Charles's law

At a constant pressure, the volume of a gas rises with its absolute temperature (in kelvin). Cool inhaled air warms in the airways and expands a little, one reason spirometry results are corrected to body temperature.

Atmospheric pressure

760 mmHg at sea level, written as 0 mmHg in respiratory physiology, so -1 mmHg means 759 mmHg.

Alveolar pressure

The air pressure inside the alveoli (intrapulmonary pressure).

Intrapleural pressure

The pressure in the pleural fluid, below atmospheric throughout quiet breathing.

Transpulmonary pressure

Alveolar minus intrapleural pressure, the difference across the lung wall that holds the lung open.

Worked example: a sealed syringe holds 10 mL of air at 760 mmHg. Pull the plunger out to 20 mL and the pressure becomes 760 × 10 ÷ 20 = 380 mmHg. With the tip open, the pressure drop pulls air in. The chest works the same way, with the breathing muscles as the hand on the plunger.

What ventilation needs

- An open airway to the alveoli.
- A sealed pleural space, so the lungs follow the chest wall.
- Working breathing muscles and the nerves that drive them.
- Lungs that stretch (compliance) and spring back (elastic recoil).
- Low airway resistance.

Pressures through one quiet breath

Point in the breath	Alveolar pressure	Intrapleural pressure	Air flow
At rest, before inspiration	0 mmHg (equal to atmospheric)	About -3 mmHg	None
Middle of inspiration	About -1 mmHg	Falling	In
End of inspiration	0 mmHg	About -6 mmHg	None; about 500 mL has entered
Middle of expiration	About +1 mmHg	Rising	Out
End of expiration	0 mmHg	About -3 mmHg	None

These are Silverthorn's round numbers. Some textbooks give a resting intrapleural pressure of about -4 mmHg (756 mmHg), and real values vary between people.

Pre-inspiration (rest between breaths)

The muscles are relaxed, alveolar pressure equals atmospheric, and no air flows. The inward pull of the lungs and outward pull of the chest wall hold intrapleural pressure at about -3 mmHg. The air in the lungs is the functional residual capacity, about 2,300 mL in a young man.

Quiet inspiration, step by step

1. Inspiratory neurons in the medulla activate motor neurons in the spinal cord, which fire through the phrenic nerves to the diaphragm and the intercostal nerves to the external intercostals.
2. The diaphragm contracts and flattens. The external intercostals lift the ribs up and out.
3. The thoracic cavity enlarges in all three directions, mostly from the diaphragm.
4. The chest wall pulls the parietal pleura out, and the pleural fluid pulls the lung along. Intrapleural pressure falls from about -3 to about -6 mmHg.
5. The lungs expand, so by Boyle's law alveolar pressure falls to about -1 mmHg.
6. Air flows down the gradient from the atmosphere through the airways into the alveoli.

7. As air enters, alveolar pressure climbs back to 0 mmHg and flow stops. About 500 mL has come in.

Quiet expiration, step by step

1. The inspiratory neurons stop firing, and the diaphragm and external intercostals relax.
2. The diaphragm springs back into its dome, and the ribs drop down and in.
3. With the muscles relaxed, elastic recoil of the stretched lungs makes the thoracic cavity smaller.
4. Alveolar volume falls, so by Boyle's law alveolar pressure rises to about +1 mmHg.
5. Air flows out of the alveoli, through the airways, to the atmosphere.
6. When alveolar pressure is back to 0 mmHg, flow stops, and intrapleural pressure is back to about -3 mmHg.

Quiet expiration is passive. No muscle contracts; the lungs empty by recoil, like a stretched rubber band let go.

Quiet versus forced breathing

	Quiet breathing	Forced breathing
Inspiration	Diaphragm and external intercostals, with some help from the scalenes	The same muscles contract harder, and the sternocleidomastoids and scalenes help lift the chest
Expiration	Passive, by elastic recoil	Active: internal intercostals pull the ribs down and in, and abdominal muscles push the diaphragm up
Pressure swings	About 1 mmHg each way in the alveoli	Much larger, so more air moves faster
When	At rest	Exercise, coughing, sneezing, playing a wind instrument, lung disease

Abdominal pressure beyond breathing

Contracting the abdominal muscles against a closed glottis (the Valsalva maneuver) raises pressure in the abdomen and chest. This helps push out feces, urine and a baby in childbirth and braces the trunk for lifting; coughs and sneezes release the same pressure suddenly. The high chest pressure also squeezes the large veins and briefly lowers venous return, so hard straining can cause lightheadedness.

The pleural seal, compliance, surfactant and airway resistance

Competencies 3, 4, 5 and 6: Intrapleural pressure; Compliance and elastic recoil; Surfactant and surface tension; Airway resistance.

In your textbook: Silverthorn Chapter 18, the parts on intrapleural pressure, lung compliance and elastance, surfactant, and airway diameter.

Dr. Rennie's videos for this part: What is a pneumothorax?, What is atelectasis?, Pulmonary compliance and surface tension, Describe airflow resistance.

Why intrapleural pressure is below atmospheric

Two forces pull in opposite directions on the sealed pleural space. The elastic lungs try to recoil inward, and the chest wall tries to spring outward. Pleural fluid cannot stretch, so the opposing pulls create a slight suction, about 3 mmHg below atmospheric at rest. This suction holds the lungs stretched open against the chest wall.

Pneumothorax, step by step

1. The pleural seal breaks, from a stab wound or broken rib, or from a weak spot on the lung surface that ruptures.
2. Air follows its pressure gradient into the pleural cavity, from the atmosphere or the alveoli.
3. Intrapleural pressure rises to atmospheric (0 mmHg).
4. Nothing holds the lung against the chest wall, so the lung recoils inward and collapses.
5. The chest wall on that side springs slightly outward.
6. The collapsed lung cannot ventilate. The other lung keeps working, because each lung has its own pleural sac.

- A spontaneous pneumothorax can happen with no injury, often in tall, thin young adults, when small air blisters (blebs) on the lung burst.
- Tension pneumothorax: damaged tissue acts as a one-way valve, so air enters with each breath but cannot leave. Pressure builds, shifts the heart and great vessels, squeezes the other lung and lowers venous return. It is an emergency.
- Removing the air, usually through a chest tube, restores the negative pressure, and the lung re-expands.
- Blood (hemothorax) or fluid (pleural effusion) in the pleural cavity can also press on the lung.

Atelectasis

Atelectasis is the collapse of alveoli in part or all of a lung, and the pleural seal can be intact. Collapsed alveoli still receive blood but no air, so that blood leaves without picking up O2.

- Blocked airway: a mucus plug, tumor or inhaled object blocks a bronchus, the trapped air is absorbed, and the alveoli beyond collapse.
- Outside pressure: a pneumothorax, pleural effusion or large tumor compresses the lung.
- Too little surfactant, as in premature newborns.
- Shallow breathing after surgery, or with pain or sedation, lets small alveoli at the lung bases close. Deep breaths, coughing, walking and an incentive spirometer reopen them.

Compliance and elastic recoil

Compliance

How easily the lungs stretch: compliance = change in volume ÷ change in pressure. High compliance means a small pressure change gives a large volume change.

Elastic recoil (elastance)

How strongly the lungs spring back after being stretched. It comes from elastin fibers and from the surface tension of the fluid lining the alveoli.

A stiff lung has low compliance and strong recoil; a floppy lung has the reverse. Healthy lungs need both to fill easily and empty passively.

	Pulmonary fibrosis	Emphysema
What happens	Scar tissue (collagen) replaces lung tissue, from dusts such as asbestos or silica, some drugs, or unknown causes	Alveolar walls and their elastin are destroyed, mostly by cigarette smoke, and alveoli merge into large air spaces
Compliance	Decreased (stiff)	Increased (floppy)
Elastic recoil	Increased	Decreased
Hardest part of breathing	Inspiration, because expanding stiff lungs takes more work	Expiration: weak recoil pushes air out poorly, and small airways that lost their elastin support collapse and trap air
Spirometry pattern	Restrictive	Obstructive
Gas exchange	A thicker membrane slows diffusion	Less surface area

Surface tension and the law of LaPlace

A thin layer of fluid lines every alveolus. At an air-water surface, water molecules pull on each other more strongly than on the air, which creates surface tension. It pulls the alveolar wall inward and resists stretching, so it makes up a large share of lung recoil.

The law of LaPlace says the pressure inside a bubble equals two times its surface tension divided by its radius: $P = 2T \div r$. In plain words, with equal surface tension the smaller bubble has the higher inward pressure and tends to collapse. Without help, small alveoli would collapse and empty into larger ones.

How surfactant fixes the problem

- Surfactant, a mix of phospholipids and proteins from type II alveolar cells, slips between the surface water molecules and lowers surface tension.
- It is more concentrated in small alveoli, so it lowers surface tension most where collapse is most likely.
- Small and large alveoli end up at similar pressures and inflate evenly.
- Compliance rises and the work of breathing falls.

Infant respiratory distress syndrome, step by step

1. Type II cells start making surfactant at about 25 weeks of gestation and usually make enough by about 34 weeks.
2. A baby born before that has too little surfactant, so alveolar surface tension is high.
3. Small alveoli collapse at the end of each expiration, and the lungs are stiff.
4. Every breath must reopen collapsed alveoli, like blowing up a new balloon each time, so the work of breathing is very high.
5. Too few alveoli stay open for gas exchange. The baby becomes hypoxic and tires.

- Signs: fast breathing, flaring nostrils, grunting, and chest wall retractions with each breath.
- Care can include artificial surfactant given into the airway and breathing support that keeps alveoli open. Before an expected early birth, the mother may get corticosteroids, which speed fetal surfactant production.
- Also called neonatal respiratory distress syndrome or hyaline membrane disease.

Airway resistance

The main determinant of airway resistance is airway radius. Resistance is proportional to 1 divided by the

radius to the fourth power (Poiseuille's law), so halving the radius raises resistance 16 times. In healthy lungs most resistance is in the nose, trachea and bronchi, because the many bronchioles side by side have a huge total cross-section. The bronchioles are where resistance changes, because their walls are smooth muscle without cartilage.

Factor	Effect on the bronchioles	Effect on airflow
Parasympathetic nerves (ACh on muscarinic receptors)	Constrict	Decreases
Histamine and leukotrienes (allergy), irritants such as smoke and dust	Constrict	Decreases
Mucus, as in chronic bronchitis and asthma	Narrows the open space	Decreases
Swelling (edema) of the airway wall	Narrows the open space	Decreases
Sympathetic stimulation: epinephrine on beta-2 receptors	Dilate	Increases
High PCO2 in the airway	Dilate	Increases

Sympathetic nerves to the bronchioles are sparse. Most sympathetic bronchodilation comes from epinephrine released by the adrenal medulla, acting on beta-2 receptors. Inhaled asthma relievers act on the same receptors.

An asthma attack, step by step

1. A trigger, such as an allergen, exercise, cold air, smoke or infection, reaches the airways of a sensitive person.
2. Mast cells and other immune cells release histamine, leukotrienes and other inflammatory chemicals.
3. Bronchiolar smooth muscle contracts (bronchospasm), the wall swells, and thick mucus is secreted.
4. Airway radius falls, so resistance rises steeply.
5. Expiration is hardest, because rising chest pressure squeezes the narrowed airways further. Air is trapped and the person wheezes.
6. FEV1 falls. The narrowing usually reverses with treatment.

Lung volumes, spirometry and ventilation

Competencies 7, 8, 9 and 10: Lung volumes and capacities; Obstructive and restrictive patterns; Alveolar ventilation and dead space; Spirometry.

In your textbook: Silverthorn Chapter 18, the parts on lung volumes and capacities and on alveolar ventilation.

Dr. Rennie's videos for this part: Spirometry, Minute ventilation, Alveolar ventilation.

A spirometer measures the air breathed in and out, and its tracing is a spirogram. There are four lung volumes. A capacity is the sum of two or more volumes.

Volume or capacity	What it is	Men	Women
Tidal volume (VT)	One quiet breath	500 mL	500 mL
Inspiratory reserve volume (IRV)	Extra air in after a quiet inspiration	3,000 mL	1,900 mL
Expiratory reserve volume (ERV)	Extra air out after a quiet expiration	1,100 mL	700 mL
Residual volume (RV)	Air left after a maximal expiration	1,200 mL	1,100 mL
Inspiratory capacity (IC) = VT + IRV	Most air in from the end of a quiet expiration	3,500 mL	2,400 mL
Functional residual capacity (FRC) = ERV + RV	Air left at the end of a quiet expiration	2,300 mL	1,800 mL
Vital capacity (VC) = IRV + VT + ERV	Most air moved in one breath	4,600 mL	3,100 mL
Total lung capacity (TLC) = VC + RV	Air in the lungs after a maximal inspiration	5,800 mL	4,200 mL

These are typical values for healthy young adults. Real values rise with height and fall with age, and women's are smaller mainly because of body size. Results are compared with predicted values from equations based on sex, age and height.

- Residual volume cannot be breathed out, so a simple spirometer cannot measure RV, FRC or TLC. Those need other methods, such as helium dilution or body plethysmography.
- Residual volume helps keep the alveoli open and lets gas exchange go on between breaths.

Forced vital capacity and FEV1

Forced vital capacity (FVC)

The volume blown out as hard and fast as possible after the deepest possible breath in.

FEV1

Forced expiratory volume in one second: the part of the FVC blown out in the first second.

FEV1/FVC ratio

The fraction of the FVC that comes out in the first second. Healthy young adults blow out about 80% of their FVC in the first second, and the ratio falls slowly with age.

	Obstructive pattern	Restrictive pattern
The problem	Air cannot get out fast, because airways are narrowed or collapse	The lungs or chest wall cannot expand fully
Examples	Asthma, COPD (chronic bronchitis and emphysema), cystic fibrosis	Pulmonary fibrosis, severe scoliosis, neuromuscular disease, severe obesity
FEV1	Decreased a lot	Decreased
FVC	Normal or decreased	Decreased
FEV1/FVC	Low, below 0.70	Normal or high
RV and TLC	Normal or increased, from trapped air	Decreased

Worked examples. Patient A: FEV1 2.0 L, FVC 4.0 L, ratio 2.0 ÷ 4.0 = 0.50, below 0.70, so obstructive. Patient B: FEV1 2.4 L, FVC 2.8 L, ratio 2.4 ÷ 2.8 = 0.86, which is normal, but the FVC is only 2.8 ÷ 4.6 × 100 = about 61% of the 4.6 L predicted for this patient, so the pattern suggests restriction.

- The 0.70 cutoff is the common simple rule. Many labs now use the lower limit of normal for the person's age, so a slightly low ratio in an older adult can be normal.
- Spirometry can suggest restriction, but confirming it requires measuring TLC.
- Asthma obstruction usually improves a lot after an inhaled bronchodilator; COPD obstruction improves less and does not fully return to normal.

Minute ventilation, alveolar ventilation and dead space

Minute ventilation

Total air moved per minute: tidal volume × breathing rate.

Anatomic dead space

The volume of the conducting zone, about 150 mL in an adult (roughly 1 mL per pound of ideal body weight). Air that stops there never reaches the alveoli.

Alveolar ventilation

Fresh air reaching the alveoli per minute: (tidal volume - dead space) × breathing rate. This is what matters for gas exchange.

Physiological dead space

Anatomic dead space plus alveoli that get air but no blood (alveolar dead space). In healthy lungs it nearly equals the anatomic dead space.

Worked example at rest: minute ventilation = 500 mL × 12 breaths per minute = 6,000 mL per minute. Alveolar ventilation = (500 - 150) mL × 12 = 4,200 mL per minute. Each breath brings only about 350 mL of fresh air into the roughly 2,300 mL already in the lungs, replacing about 15% of alveolar air, which keeps alveolar PO2 and PCO2 steady between breaths.

Breathing pattern	Tidal volume × rate	Minute ventilation	Alveolar ventilation
Normal	500 mL × 12	6,000 mL/min	(500 - 150) × 12 = 4,200 mL/min
Rapid and shallow	250 mL × 24	6,000 mL/min	(250 - 150) × 24 = 2,400 mL/min
Slow and deep	750 mL × 8	6,000 mL/min	(750 - 150) × 8 = 4,800 mL/min

All three move 6,000 mL per minute, but the dead space takes 150 mL out of every breath, so more breaths means more wasted air. Slow deep breathing ventilates the alveoli best. Rapid shallow breathing, for example from pain after chest or abdominal surgery, can underventilate the alveoli even when minute ventilation looks normal.

Spirometry lab, step by step

1. Sit upright, wear the nose clip, and seal your lips around a clean mouthpiece.
2. Breathe quietly until the tracing is steady. Each quiet breath is the tidal volume.
3. After a quiet inspiration, breathe in as deeply as you can. The extra rise is the IRV.
4. After a quiet expiration, breathe out as far as you can. The extra drop is the ERV.
5. For the forced test, breathe in fully, then blow out as hard and fast as you can until no more air comes out (at least 6 seconds for an adult). Do at least three tries and use the largest FVC and FEV1.
6. Calculate VC = IRV + VT + ERV, IC = VT + IRV, and FEV1/FVC.
7. Compare with predicted values: percent predicted = measured ÷ predicted × 100. About 80% or higher has traditionally been read as normal.

- RV, FRC and TLC cannot come from this tracing. A TLC made with a typical RV is only an estimate.

- Common errors: weak effort, air leaking around the mouthpiece, coughing, or not breathing in fully first.

Gas exchange: partial pressures, diffusion and ventilation-perfusion matching

Competencies 11, 12 and 13: Partial pressures; Diffusion at the respiratory membrane; Ventilation perfusion matching.

In your textbook: Silverthorn Chapter 19, the part on gas exchange in the lungs and tissues, and Chapter 18, the part on matching ventilation to alveolar blood flow.

Dr. Rennie's videos for this part: Gas exchange, Dalton's law, PO2, PCO2, Henry's law, Fick's law of diffusion, Diffusion time, Ventilation-perfusion coupling, part 1, Ventilation-perfusion coupling, part 2, Ventilation-perfusion coupling, part 3, Ventilation-perfusion coupling, part 4.

Dalton's law and partial pressure

Dalton's law says the total pressure of a gas mixture is the sum of the pressures of each gas in it. Partial pressure = total pressure × the fraction of the mixture that gas makes up. Dry air is about 78% N2, 21% O2 and 0.04% CO2. Each gas moves down its own partial pressure gradient, whatever the other gases do.

Worked calculation: PO2 from the air to the alveoli

1. Dry air at sea level: PO2 = 760 mmHg × 0.21 = about 160 mmHg.
2. Inhaled air is humidified. Water vapor at 37 °C (98.6 °F) has a partial pressure of 47 mmHg, which takes up part of the total.
3. Humidified inspired air: PO2 = (760 - 47) × 0.21 = 713 × 0.21 = about 150 mmHg.
4. In the alveoli, O2 keeps diffusing into the blood and CO2 keeps arriving from it, and fresh air mixes with the air already there. Alveolar PO2 settles at about 100 mmHg and PCO2 at about 40 mmHg.

Location	PO2 (mmHg)	PCO2 (mmHg)
Dry air at sea level	160	0.3
Humidified air in the trachea	150	Almost 0
Alveolar air	100	40
Arterial blood	About 95 (normal 80 to 100)	40 (normal 35 to 45)
Resting tissue cells	40 or less	46 or more
Venous blood leaving resting tissue	40	46

Arterial PO2 is a little below alveolar PO2 because some venous blood, mostly from the bronchial veins draining the lung tissue, joins the pulmonary veins, and ventilation and blood flow are not perfectly matched everywhere. Arterial PO2 also drifts down with age.

Gas exchange at the lungs and tissues, step by step

1. Venous blood (PO2 40, PCO2 46 mmHg) reaches the pulmonary capillaries.
2. O2 diffuses from the alveoli (100) into the blood; CO2 diffuses from the blood (46) into the alveoli (40).
3. Blood leaving the alveoli matches alveolar air (PO2 100, PCO2 40 mmHg). After mixing with a little venous blood, it enters the systemic arteries at a PO2 of about 95 and a PCO2 of 40.
4. In the systemic capillaries, O2 diffuses into the cells and CO2 diffuses out of them into the blood.
5. Blood leaves the tissues at a PO2 of 40 and a PCO2 of 46 and returns to the right heart.

The CO2 gradient at the lung (6 mmHg) is ten times smaller than the O2 gradient (60 mmHg), yet CO2 exchanges just as well, because it is far more soluble.

Henry's law and Fick's law

Henry's law

The amount of a gas that dissolves in a liquid is proportional to its partial pressure and its solubility. CO2 is about 20 times more soluble in water than O2. At a PO2 of 100 mmHg only about 0.3 mL of O2 dissolves in 100 mL of blood, far too little for the body, which is why hemoglobin is needed.

Fick's law of diffusion

Diffusion across a membrane is faster with more surface area, a bigger partial pressure gradient and a

more soluble gas, and slower with a thicker membrane.

Factor	Normal lung	What lowers it	Examples
Surface area	Roughly 70 square meters	Fewer alveoli	Emphysema, removal of part of a lung
Partial pressure gradient	Alveolar PO_2 100 versus venous 40 mmHg	Lower alveolar PO_2	High altitude, hypoventilation
Membrane thickness	Less than 1 µm	Thicker barrier or fluid in the way	Fibrosis, pulmonary edema, pneumonia
Solubility	CO_2 about 20 times O_2	Fixed for each gas	O_2 transfer suffers first

Because CO_2 diffuses so much more easily, a diffusion problem shows up first as low arterial PO_2 (hypoxemia) with a normal or even low PCO_2. The low O_2 makes the person breathe harder, which blows off CO_2.

Pulmonary edema, step by step

1. Pulmonary capillary pressure rises, for example when a failing left ventricle lets blood back up into the lungs.
2. More fluid filters into the interstitial space between capillaries and alveoli, and the diffusion distance increases.
3. In severe edema, fluid floods the alveoli themselves.
4. O_2 transfer falls first, and arterial PO_2 drops.

Diffusion time

At rest a red blood cell spends about 0.75 second in a pulmonary capillary, but its PO_2 matches alveolar air within about the first 0.25 second, a wide safety margin. In heavy exercise the trip can shorten to about 0.25 second. Healthy lungs still finish loading, but with a thickened membrane, as in fibrosis, arterial PO_2 can be normal at rest and fall during exertion.

Matching ventilation to perfusion

Gas exchange works best when the air reaching each group of alveoli matches the blood flowing past them. For the whole lung at rest, alveolar ventilation is about 4 L per minute and pulmonary blood flow about 5 L per minute, a ventilation-perfusion ratio (V/Q) of about 0.8.

Standing up, both ventilation and blood flow are greater at the bottom of the lungs, but blood flow increases more. The top has more air than blood (high V/Q), and the base more blood than air (low V/Q). Local responses fine-tune the match region by region.

Local change in the alveoli	Response	Result
PO_2 falls (too little air for the blood)	Pulmonary arterioles supplying them constrict	Blood goes to better-ventilated alveoli
PO_2 rises	Pulmonary arterioles dilate	More blood to well-ventilated alveoli
PCO_2 rises (too little air for the blood)	Bronchioles serving them dilate	More air arrives to carry off the CO_2
PCO_2 falls (too much air for the blood)	Bronchioles constrict	Air goes to alveoli with more blood flow

Pulmonary arterioles respond to low O_2 the opposite way from systemic arterioles, which dilate to bring blood to a tissue short of O_2. Pulmonary arterioles constrict (hypoxic pulmonary vasoconstriction), so blood avoids alveoli where it cannot load O_2.

A partly blocked airway, step by step

1. Mucus partly blocks a bronchiole, and ventilation to the alveoli beyond falls.
2. Their alveolar PO_2 falls and their PCO_2 rises.
3. Low PO_2 constricts the local pulmonary arterioles, sending blood to well-ventilated alveoli.
4. High PCO_2 dilates the local bronchioles, letting in more air past the partial block.
5. The match improves, and arterial PO_2 falls less than it otherwise would.

When the match fails

Shunt (V/Q near 0)

Blood flows past alveoli that get no air, as in pneumonia or atelectasis, and that O_2–poor blood causes hypoxemia. Extra O_2 helps little with a true shunt, because the shunted blood never meets it.

Alveolar dead space (V/Q very high)

Alveoli get air but no blood, as when a pulmonary embolism blocks a pulmonary artery.

Widespread low O_2

When the whole lung is hypoxic (severe COPD, altitude), pulmonary arterioles constrict everywhere, causing pulmonary hypertension, and over time the right ventricle can fail (cor pulmonale).

Oxygen transport and the oxyhemoglobin curve

Competencies 14, 15 and 17: Oxygen transport and the oxyhemoglobin curve; Curve shifts; Bohr and Haldane effects.

In your textbook: Silverthorn Chapter 19, the part on gas transport in the blood, including oxygen binding to hemoglobin and the oxyhemoglobin dissociation curve.

Dr. Rennie's videos for this part: Gas transport, O2–Hb dissociation curve, part 1, O2–Hb dissociation curve, part 2, O2 loading and unloading, O2 loading and unloading of fetal hemoglobin, O2 loading and unloading during exercise.

More than 98% of the O2 in blood is bound to hemoglobin (Hb) inside red blood cells, and less than 2% is dissolved in plasma. Each Hb has four heme groups, and the iron in each heme binds one O2, so one Hb carries up to four O2.

Hb and O2 form oxyhemoglobin (HbO2), and the reaction runs both ways. By the law of mass action, high PO2 in the lungs pushes O2 onto Hb (loading), and low PO2 in the tissues lets it come off (unloading). Only dissolved O2 creates PO2, so Hb soaking up O2 in the lungs keeps plasma PO2 low and lets more O2 diffuse in.

Saturation

The percent of hemoglobin's O2 binding sites holding O2. Arterial saturation (SaO2) is normally about 95 to 100%. A pulse oximeter estimates it through the skin (SpO2).

Oxyhemoglobin dissociation curve

A graph of hemoglobin saturation against PO2. It is S-shaped (sigmoid).

P50

The PO2 at which hemoglobin is 50% saturated, about 27 mmHg for adult hemoglobin. A higher P50 means a right shift and weaker binding.

PO2 (mmHg)	Saturation	Where you find it
100	About 98%	Alveolar air and arterial blood
60	About 90%	Lower edge of the flat part
40	About 75%	Venous blood leaving resting tissue
27	50%	The P50
20	About 30 to 35%	Venous blood leaving hard-working muscle

Why the curve is S-shaped

- At very low PO2, deoxyhemoglobin binds O2 weakly, so the curve starts slowly.
- Once one heme binds O2, the molecule changes shape and the other hemes bind O2 more easily (cooperative binding), so the middle of the curve is steep.
- As the last sites fill, few are left free, and the curve levels into a plateau.

Part of the curve	PO2 range	Advantage
Flat upper part	About 60 to 100 mmHg	Safety margin for loading: alveolar PO2 can fall a lot, at altitude or in mild lung disease, and Hb stays about 90% saturated
Steep lower part	About 10 to 40 mmHg	Unloading: a small drop in tissue PO2 releases a lot of O2, so busier tissues automatically get more

At rest, blood gives up only about 25% of its O2, falling from about 98% to 75% saturation, and the rest is a venous reserve. In hard-working muscle, PO2 can fall to 20 mmHg or lower, saturation to about 30 to 35%, and the blood gives up two-thirds or more of its O2.

Shifts of the curve

A right shift means weaker binding: at any PO2, Hb holds less O2, so more is released to the tissues. A left shift means tighter binding: Hb loads easily but releases less.

Change	Shift	Hemoglobin's hold on O2	Effect on the tissues
H+ rises (pH falls)	Right	Weaker	More O2 unloaded
PCO2 rises	Right	Weaker	More O2 unloaded
Temperature rises	Right	Weaker	More O2 unloaded
2,3-BPG rises	Right	Weaker	More O2 unloaded
pH rises, PCO2 falls, cooler, or less 2,3-BPG	Left	Stronger	Less O2 unloaded
Carbon monoxide on some hemes; fetal hemoglobin	Left	Stronger	Less O2 unloaded

Bohr effect

The right shift caused by H+ and CO2. H+ binds hemoglobin and changes its shape, lowering its O2 affinity. CO2 does the same by forming H+ and by binding hemoglobin directly. In the tissues this helps unload O2. In the lungs, as CO2 is breathed off and pH rises, affinity rises and loading improves.

2,3-bisphosphoglycerate (2,3-BPG)

Made by red blood cells during glycolysis. It binds deoxyhemoglobin and lowers its O2 affinity. It rises over a day or two of ongoing hypoxia (high altitude, anemia, chronic lung disease).

Loading and unloading during exercise

Working muscle is low in O2 and makes CO2, H+ and heat, so blood there sits on the steep part of a right-shifted curve and releases far more O2. In the lungs, with lower PCO2 and cooler blood, the curve shifts back and loading stays near normal.

Fetal hemoglobin

Fetal hemoglobin (HbF) has two gamma chains in place of the two beta chains of adult hemoglobin. It binds 2,3-BPG weakly, so its O2 affinity is higher and its curve lies left of the mother's. In the placenta, where PO2 is low, fetal blood can still load the O2 that maternal hemoglobin unloads, and fetal CO2 entering maternal blood shifts the mother's curve right. Adult hemoglobin replaces most HbF in the first months after birth.

Carbon dioxide transport, the Haldane effect and oxygen content

Competencies 16, 17 and 18: Carbon dioxide transport; Bohr and Haldane effects; Oxygen content versus partial pressure.

In your textbook: Silverthorn Chapter 19, the parts on carbon dioxide transport and on oxygen transport in the blood.

Dr. Rennie's videos for this part: Transport of CO2 (CO2 loading).

At rest the body makes about 200 mL of CO2 per minute. It must be removed, because CO2 forms acid (H+) in body fluids, and very high levels depress the nervous system.

Form	Share of CO2 carried	Where
Dissolved CO2	About 7%	Plasma and red cell fluid
Bound to hemoglobin (carbaminohemoglobin)	About 23%	On amino groups of the globin chains, not the iron
Bicarbonate (HCO3−)	About 70%	Made in red blood cells, carried mostly in plasma

These are Silverthorn's figures; other textbooks give slightly different splits, such as 10, 20 and 70%. CO2 binds globin, not heme, so it does not compete with O2 for the same site.

The key reaction: CO2 + H2O form carbonic acid (H2CO3), which splits into H+ and HCO3−, and it runs both ways. It is slow in plasma and very fast in red cells, where the enzyme carbonic anhydrase speeds it.

CO2 loading in a systemic capillary, step by step

1. Tissue PCO2 (46 mmHg or more) is above that of arriving blood (40 mmHg), so CO2 diffuses from the cells into the plasma.
2. A little stays dissolved in plasma. Most diffuses on into the red blood cells.
3. Some binds the globin chains as carbaminohemoglobin.
4. Most combines with water, helped by carbonic anhydrase, to form carbonic acid, which splits into H+ and HCO3−.
5. Hemoglobin binds and buffers the H+. H+ binding also lowers hemoglobin's O2 affinity (the Bohr effect), so more O2 goes to the tissue.
6. HCO3− leaves the red cell through a bicarbonate-chloride exchanger (band 3 protein) as Cl- moves in. This chloride shift keeps the charges balanced.
7. The HCO3− travels in the plasma to the lungs.

CO2 unloading in a pulmonary capillary, step by step

1. Venous blood arrives at a PCO2 of 46 mmHg, above alveolar PCO2 (40 mmHg), so dissolved CO2 diffuses into the alveoli.
2. As plasma PCO2 falls, CO2 diffuses out of the red cells and on into the alveoli.

3. O2 binds hemoglobin, which then releases its H+ and bound CO2 (the Haldane effect).
4. HCO3– moves back into the red cell as Cl- moves out: the chloride shift in reverse.
5. HCO3– joins the released H+ to form carbonic acid, and carbonic anhydrase turns it into CO2 and water.
6. That CO2 diffuses into the alveoli and is breathed out.

Arterial blood still carries a large amount of CO2, mostly as HCO3–. This bicarbonate is the main buffer in plasma and the other extracellular fluid, normally about 22 to 26 mEq/L in plasma.

The Bohr and Haldane effects together

	Bohr effect	Haldane effect
The rule	More CO2 and H+ means weaker O2 binding (right shift)	Less O2 bound means more CO2 and H+ bound
At the tissues	CO2 and H+ from the cells help hemoglobin release O2	As O2 leaves, deoxyhemoglobin picks up more CO2 and H+
At the lungs	As CO2 is breathed off and pH rises, O2 binds more strongly	As O2 binds, hemoglobin releases CO2 and H+

Both effects come from the same shape change in hemoglobin. At each site, moving one gas makes it easier to move the other gas the way it needs to go.

Oxygen content versus PO2 and saturation

PO2

The pressure of O2 dissolved in plasma. It drives diffusion and sets saturation, but it does not tell you how much O2 the blood holds.

Saturation

The percent of hemoglobin sites holding O2. It does not tell you how much hemoglobin there is.

O2 content

Total O2 in the blood, bound plus dissolved, in mL of O2 per 100 mL of blood (mL/dL). Normal arterial blood holds about 20 mL/dL.

O2 content = (Hb in g/dL × 1.34 × saturation) + (0.003 × PO2). Each gram of fully saturated hemoglobin carries about 1.34 mL of O2, and each mmHg of PO2 dissolves about 0.003 mL of O2 in 100 mL of blood.

	Healthy	Anemia	Carbon monoxide poisoning
Hemoglobin	15 g/dL	7.5 g/dL	15 g/dL
Arterial PO2	95 mmHg	95 mmHg	95 mmHg
Sites holding O2	98%	98%	About 50%, with CO on most of the rest
O2 content	15 × 1.34 × 0.98 + 0.3 = about 20 mL/dL	7.5 × 1.34 × 0.98 + 0.3 = about 10 mL/dL	15 × 1.34 × 0.50 + 0.3 = about 10 mL/dL
Unloading in the tissues	Normal	Normal curve, but less O2 to give	Worse: CO shifts the curve left, so the bound O2 is held tightly

In anemia, PO2, saturation and the pulse oximeter reading are normal; the blood just has half the hemoglobin and carries about half the O2.

Carbon monoxide binds heme more than 200 times as tightly as O2, so even a little in the air takes over many sites. PO2 stays normal, so the peripheral chemoreceptors are barely stimulated and breathing may not speed up. A standard pulse oximeter cannot tell CO-bound from O2–bound hemoglobin, so SpO2 can read falsely normal; CO-oximetry on a blood sample gives the true value. High-concentration O2 speeds the release of CO.

- Healthy arterial blood holds about 20 mL/dL and venous blood from resting tissue (75% saturated) about 15 mL/dL, so resting tissues take about 5 mL/dL, the 25% extraction described earlier.
- Extra O2 adds very little content when saturation is already 98%; it mostly raises the small dissolved amount.

Control of breathing: respiratory centers and chemoreceptors

Competencies 19 and 20: Respiratory centers; Chemoreceptor control.

In your textbook: Silverthorn Chapter 19, the part on the regulation of ventilation.

Dr. Rennie's videos for this part: Neural control of breathing, Respiratory centers of the brainstem, part 1, Respiratory centers of the brainstem, part 2, Central and peripheral inputs to the respiratory system, part 1, Central and peripheral inputs to the respiratory system, part 2,

Central and peripheral inputs to the respiratory system, part 3, Central and peripheral inputs to the respiratory system, part 4, Chemoreceptors and gas levels (CO2 and O2, the primary stimulus to breathe), Voluntary control of breathing.

Each automatic breath starts as a burst of action potentials from brainstem neurons. Sensory inputs adjust it, and the cortex can override it for a while.

The respiratory centers

Center	Location	Role
Dorsal respiratory group (DRG)	Medulla, in the nucleus tractus solitarius	Mostly inspiratory neurons driving the diaphragm through the phrenic nerves. Receives chemoreceptor and stretch receptor input through cranial nerves IX and X.
Ventral respiratory group (VRG)	Medulla, in front of and beside the DRG	Neurons for inspiration and active expiration; drives accessory and expiratory muscles in forced breathing and the muscles that keep the throat open. Contains the pre-Bötzinger complex, the rhythm generator.
Pontine respiratory group (PRG), formerly the pneumotaxic center	Upper pons	Fine-tunes the rhythm, helps end each inspiration, and smooths the switch to expiration.
Apneustic center	Lower pons	An older term for an area that promotes inspiration; whether it is a separate center in humans is uncertain.

The older names come from animal experiments. The current view is that a neuron network in the medulla generates the rhythm and the pons modifies it.

How the rhythm is generated, step by step

1. Pacemaker-like neurons in the pre-Bötzinger complex of the medulla fire on their own.
2. They switch on inspiratory neurons in the DRG and VRG, whose firing ramps up over about 2 seconds.
3. The ramp recruits more motor neurons in the phrenic and intercostal nerves, so the diaphragm and external intercostals contract smoothly and inspiration begins.
4. The inspiratory neurons shut off abruptly, helped by input from the pons and from lung stretch receptors.
5. The muscles relax, and quiet expiration happens passively over about 3 seconds.
6. The cycle repeats about 12 to 20 times per minute in an adult at rest.

Inputs that modify the rhythm

Pulmonary stretch receptors

In airway smooth muscle. When the lungs are well inflated, they signal through the vagus nerve to end inspiration (the Hering-Breuer inflation reflex). In adults this acts mainly at large tidal volumes, as in exercise; it matters more in infants.

Irritant receptors

In the airway lining. Dust, smoke, cold air and chemicals trigger coughing or sneezing and can constrict the bronchioles.

Proprioceptors

In moving muscles and joints; they help raise breathing when exercise starts.

Higher brain centers

The cortex gives voluntary control for speaking, singing and breath holding. The limbic system and hypothalamus change breathing with emotion, pain and body temperature (crying, fear, anxiety, fever).

Voluntary control and its limit

You can hold your breath or breathe fast on purpose because the cortex signals the respiratory motor neurons in the spinal cord directly, bypassing the brainstem. The control is temporary: during a breath hold, CO2 builds until the chemoreceptor drive wins at the breaking point.

Central and peripheral chemoreceptors

	Central chemoreceptors	Peripheral chemoreceptors
Location	Surface of the medulla, bathed in cerebrospinal fluid (CSF)	Carotid bodies where each common carotid artery splits; aortic bodies on the aortic arch
Nerve to the centers	None needed; they sit in the brainstem	Glossopharyngeal nerve (IX) from the carotid bodies; vagus nerve (X) from the aortic bodies
Main stimulus	H+ in the CSF, formed from arterial CO2	Arterial PO2 below about 60 mmHg, rising PCO2, rising plasma H+
Respond to low O2?	No	Yes; the only O2 sensors that drive breathing
Respond to H+ from metabolic acids?	Little, because plasma H+ crosses the blood-brain barrier poorly	Yes, mainly the carotid bodies

The central chemoreceptor response to CO2, step by step

1. Arterial PCO2 rises, for example during a breath hold.
2. CO2 crosses the blood-brain barrier into the CSF and forms carbonic acid, which releases H+.
3. The H+ stimulates the central chemoreceptors, which signal the respiratory centers.
4. The centers increase the rate and depth of breathing.
5. More CO2 is breathed off, and arterial PCO2 falls back toward 40 mmHg (negative feedback).

The carotid body response to low O2, step by step

1. Arterial PO2 falls below about 60 mmHg.
2. In the glomus cells of the carotid body, O2–sensitive K+ channels close.
3. With less K+ leaving, the glomus cell depolarizes.
4. Voltage-gated Ca2+ channels open and Ca2+ enters.
5. The cell releases neurotransmitter onto sensory neurons of the glossopharyngeal nerve.
6. The signal reaches the respiratory centers in the medulla, which increase ventilation.

Why CO2 is the main drive to breathe

- Arterial PCO2 is held near 40 mmHg breath by breath, and a rise of a few mmHg produces a large increase in ventilation.
- Low O2 adds little drive until PO2 falls below about 60 mmHg, because above that hemoglobin is still about 90% saturated.
- CSF has much less protein to buffer H+ than blood, so a small rise in CO2 gives a relatively large rise in CSF H+.
- In some people with severe COPD who retain CO2 long term, CSF bicarbonate rises and the central chemoreceptors adapt, so the response to low O2 matters relatively more, and O2 is given carefully to a target saturation. Experts now think the extra rise in PCO2 these patients can have on too much O2 comes mainly from worse V/Q matching and the Haldane effect, more than from losing a hypoxic drive.

Exercise, altitude, the ventilation lab and respiratory terms

Competencies 21 and 22: Ventilation in exercise and at altitude; Ventilatory response testing.

In your textbook: Silverthorn Chapter 19, the parts on gas exchange and the regulation of ventilation; Chapter 25, Integrative Physiology III: Exercise, the part on ventilatory responses to exercise.

Dr. Rennie's videos for this part: Respiratory terminology: dyspnea, eupnea and more, Respiratory terminology: hypercarbia, hypercapnia and more, Respiratory terminology: respiratory alkalosis and acidosis, Hypoxia.

Ventilation in exercise, step by step

1. Within seconds of starting, before blood gases change, ventilation jumps, driven by feedforward signals from the motor cortex and from proprioceptors in moving muscles and joints.
2. Over a few minutes it rises gradually to a steady level matching the O2 used and CO2 made. Tidal volume rises first, then rate.
3. In moderate exercise, arterial PO2, PCO2 and pH stay close to normal. This is hyperpnea, not hyperventilation, so the chemoreceptors are not the main trigger.
4. In hard exercise, above the ventilatory threshold, lactic acid adds H+. The carotid bodies respond, ventilation outpaces CO2 production, and arterial PCO2 falls.
5. When exercise stops, ventilation drops quickly, then returns slowly to resting while the body uses extra O2 to recover.

Minute ventilation can rise from about 6 L per minute at rest to more than 100 L per minute in fit adults. Exactly which signals match it so closely to exercise is still debated.

Altitude

Air at altitude is still 21% O2, but total pressure is lower. At about 5,500 m (18,000 ft), barometric pressure is about half of sea level, about 380 mmHg. Dry-air PO2 = 380 × 0.21 = about 80 mmHg. Humidified, PO2 = (380 - 47) × 0.21 = about 70 mmHg, and alveolar PO2 is lower still.

	Acute exposure (hours)	Chronic exposure (days to weeks)
Ventilation	Peripheral chemoreceptors sense low PO2, and ventilation rises	Rises further over several days (ventilatory acclimatization)
PCO2 and pH	Extra breathing blows off CO2, and respiratory alkalosis develops, which holds back the rise in breathing	The kidneys excrete more HCO3−, pH moves back toward normal, and that brake is released
Hemoglobin curve	Alkalosis shifts it left	2,3–BPG rises within days and shifts it right, helping unloading
Red blood cells	No change yet	The kidneys release more erythropoietin (EPO), and red cell count and hematocrit rise over weeks
Pulmonary circulation	Hypoxic vasoconstriction throughout the lungs raises pulmonary artery pressure	Pulmonary hypertension can persist

- Heart rate and cardiac output rise on arrival.
- Acute mountain sickness brings headache, nausea, fatigue and poor sleep. Serious forms are high-altitude pulmonary edema (HAPE) and high-altitude cerebral edema (HACE). Going down is the most important treatment.

Ventilatory response lab, step by step

1. Record resting rate and depth (or tidal volume) for one minute as the baseline.
2. Do one maneuver: hyperventilation, breath holding, or rebreathing into a bag.
3. Record rate and depth until breathing returns to baseline.
4. Explain each result by what happened to CO2.

Test	What happens to CO2	Breathing afterward	Why
Hyperventilation (fast, deep breathing for about a minute)	PCO2 falls (hypocapnia)	Slower and shallower, sometimes a brief pause (apnea)	Low CO2 removes the chemoreceptor drive until CO2 builds back up
Breath holding	PCO2 rises and PO2 falls	Faster and deeper	Rising CO2 stimulates the chemoreceptors and mainly sets the breaking point; falling O2 adds to it
Breath holding after hyperventilation	Starts lower, so it takes longer to reach the breaking point	Longer hold, then faster and deeper breathing	The drive starts from a lower CO2
Rebreathing from a bag	PCO2 rises steadily; PO2 falls	Depth, then rate, increase steadily	Rising CO2 is the main stimulus

- Lightheadedness and tingling in fingers and lips during hyperventilation come from low CO2, which constricts brain arterioles, and from the alkalosis, which lowers ionized Ca2+ and makes nerves more excitable.
- Never hyperventilate before swimming underwater. Low CO2 delays the urge to breathe, so PO2 can fall far enough to cause fainting first (shallow water blackout).
- With a CO2 absorber in the rebreathing bag, CO2 stays near normal while O2 falls, and breathing rises far less until PO2 gets low, which shows that CO2 is the main drive.
- Stop any test and sit down if you feel dizzy.

Respiratory terms

Eupnea

Normal, quiet breathing at rest.

Hyperpnea

Deeper, faster breathing that matches a higher metabolic rate, as in exercise. Arterial PCO2 stays about normal.

Hyperventilation

Breathing more than metabolism requires, so arterial PCO_2 falls.

Hypoventilation

Breathing less than metabolism requires, so arterial PCO_2 rises.

Tachypnea

Fast breathing, often shallow; in an adult, more than about 20 breaths per minute.

Bradypnea

Abnormally slow breathing; in an adult, fewer than about 12 breaths per minute.

Apnea

No breathing, as in a breath hold or the pauses of sleep apnea.

Dyspnea

The feeling of difficult breathing or not getting enough air (shortness of breath).

Orthopnea

Dyspnea when lying flat that eases on sitting up, common in heart failure.

Hypercapnia (hypercarbia)

Arterial PCO_2 above 45 mmHg.

Hypocapnia (hypocarbia)

Arterial PCO_2 below 35 mmHg.

Hypoxemia

Low arterial PO_2, below about 80 mmHg.

Hypoxia

Too little O_2 reaching or being used by the tissues.

Cyanosis

A bluish color of the skin and mucous membranes when much of the hemoglobin is deoxygenated.

Respiratory acidosis and alkalosis

pH falls when hypoventilation retains CO_2 (acidosis) and rises when hyperventilation removes CO_2 (alkalosis). Buffers, compensation and all four acid-base disorders are in the Acid-base balance notes.

Type of hypoxia	The problem	Examples
Hypoxic (hypoxemic) hypoxia	Low arterial PO_2	High altitude, hypoventilation, diffusion problems, V/Q mismatch
Anemic hypoxia	Too little O_2 carried, with normal PO_2	Anemia, carbon monoxide poisoning
Ischemic (circulatory or stagnant) hypoxia	Too little blood flow	Heart failure, shock, a blocked artery
Histotoxic hypoxia	Cells cannot use the O_2 delivered	Cyanide poisoning